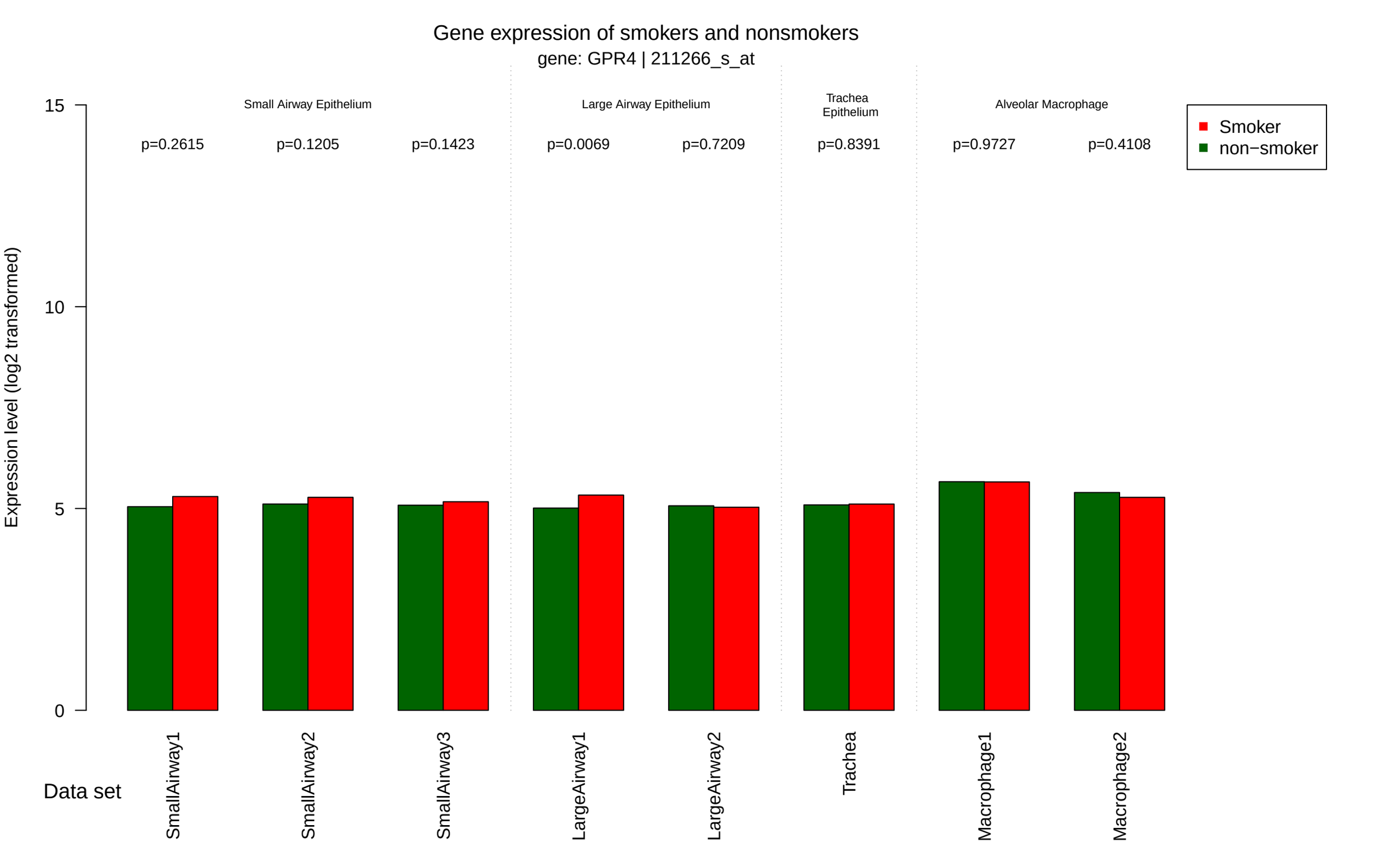
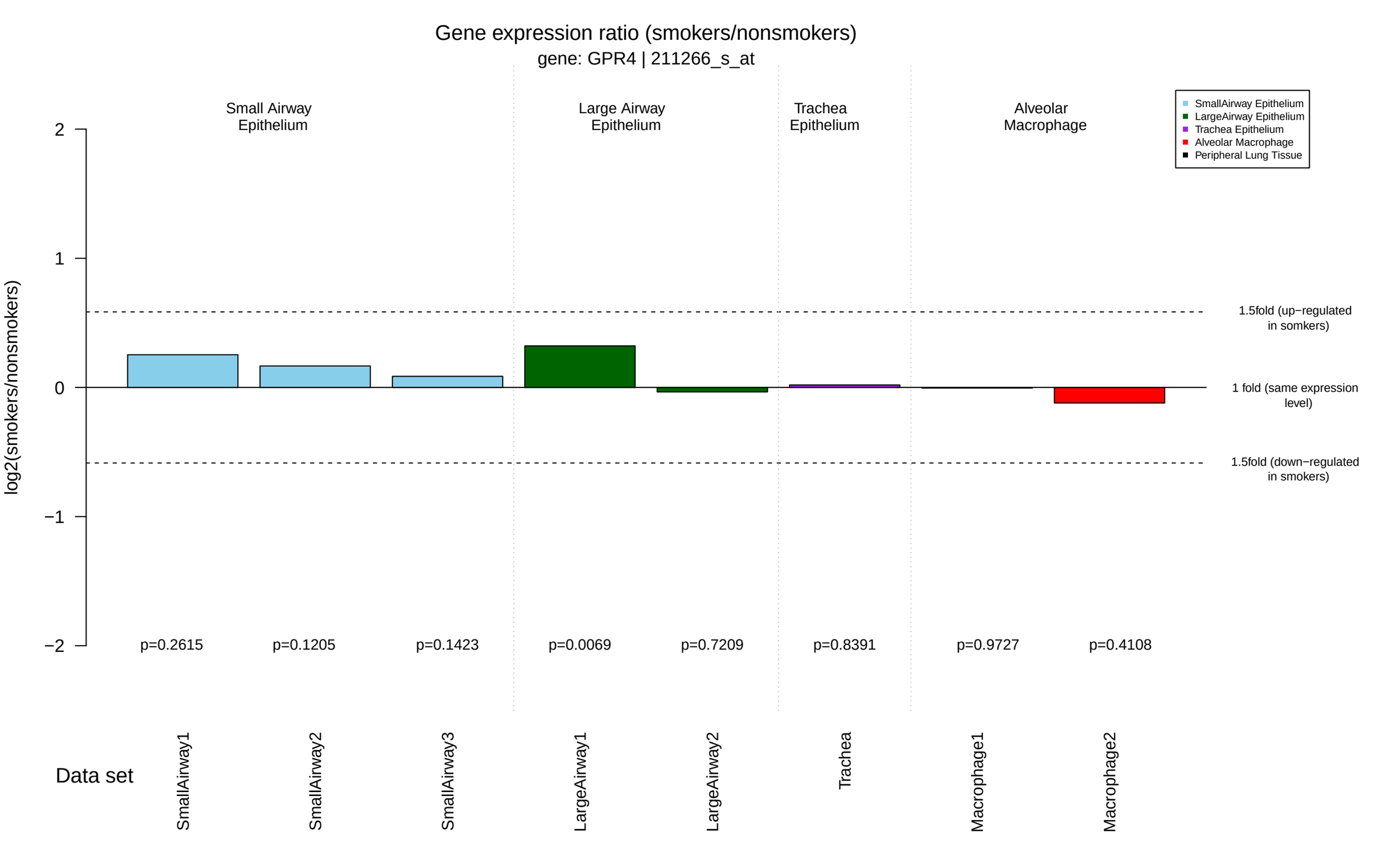
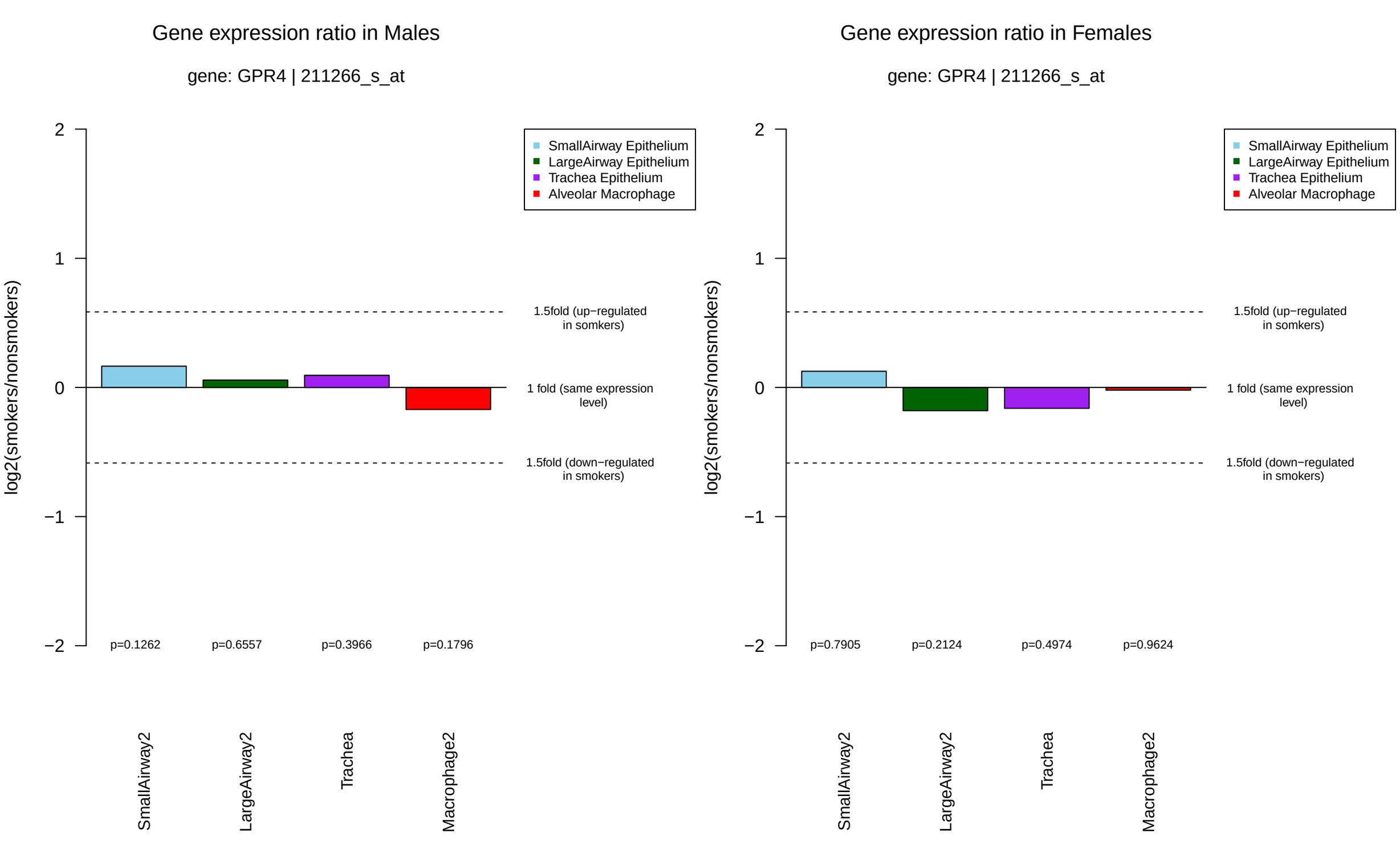
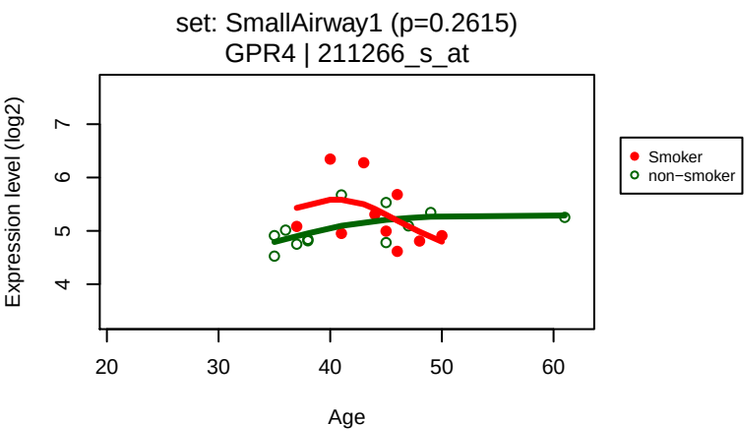
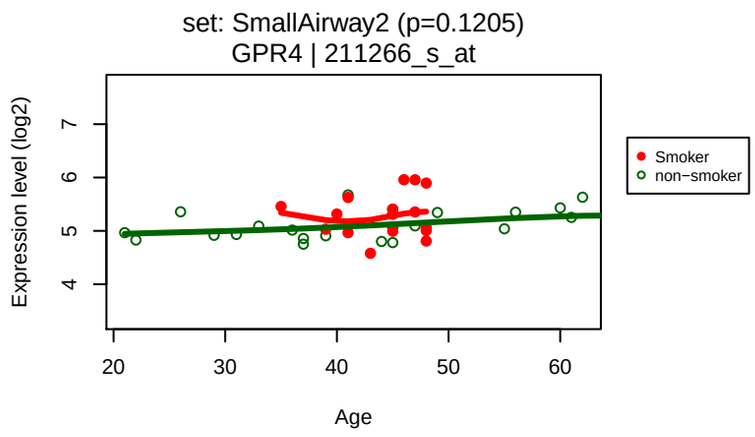
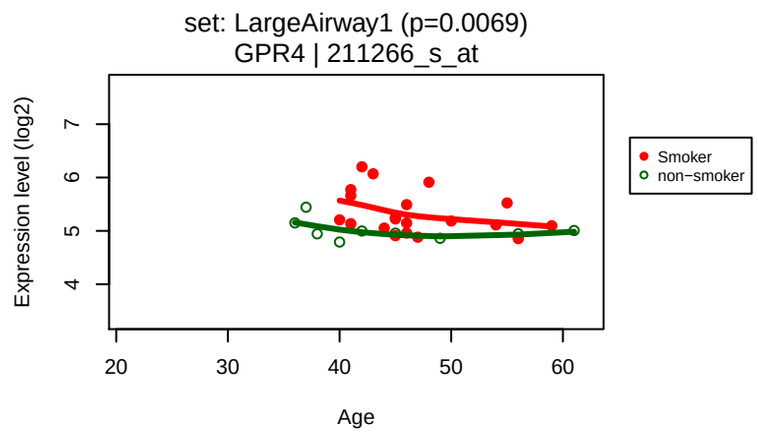

Gene expression of smokers and nonsmokers
gene: GPR4 | 211266_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2615
p=0.1205
p=0.1423
p=0.0069
p=0.7209
p=0.8391
p=0.9727
p=0.4108
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

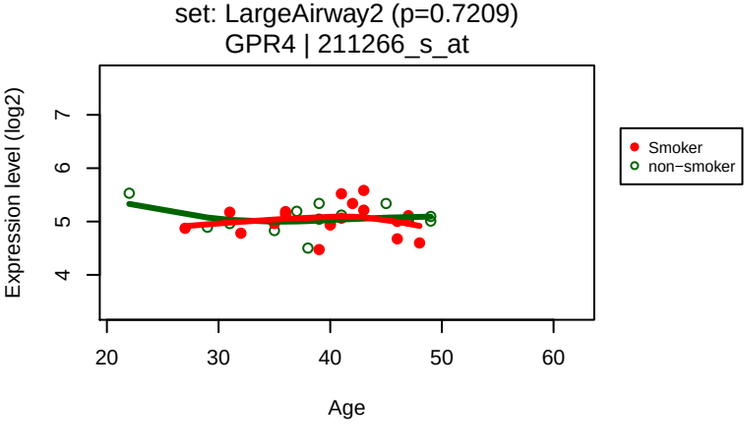

Gene expression ratio (smokers/nonsmokers)
gene: GPR4 | 211266_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2615
p=0.1205
p=0.1423
p=0.0069
p=0.7209
p=0.8391
p=0.9727
p=0.4108
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

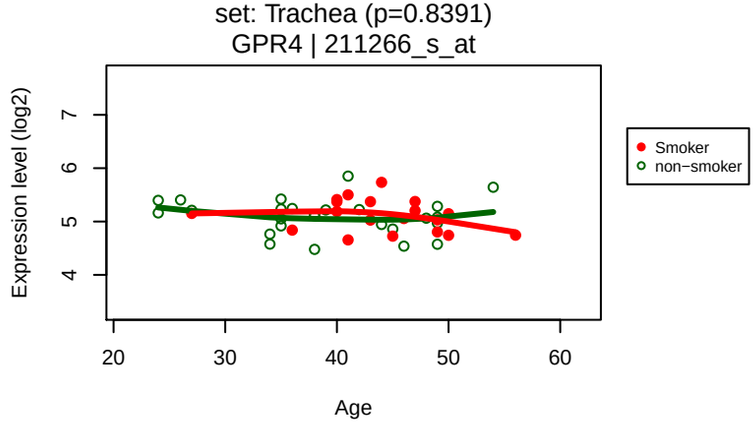

Gene expression ratio in Males
gene: GPR4 | 211266_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1262
p=0.6557
p=0.3966
p=0.1796
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GPR4 | 211266_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7905
p=0.2124
p=0.4974
p=0.9624
SmallAirway2
LargeAirway2
Trachea
Macrophage2

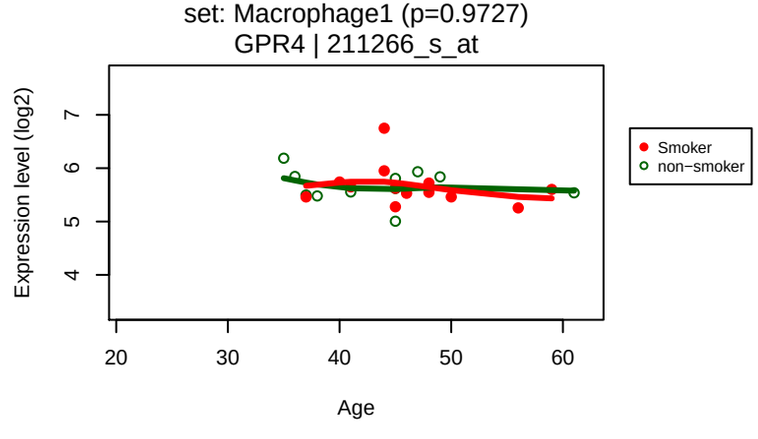

set: SmallAirway1 (p=0.2615)
GPR4 | 211266_s_at
Expression level (log2)
Age
Smoker
non−smoker

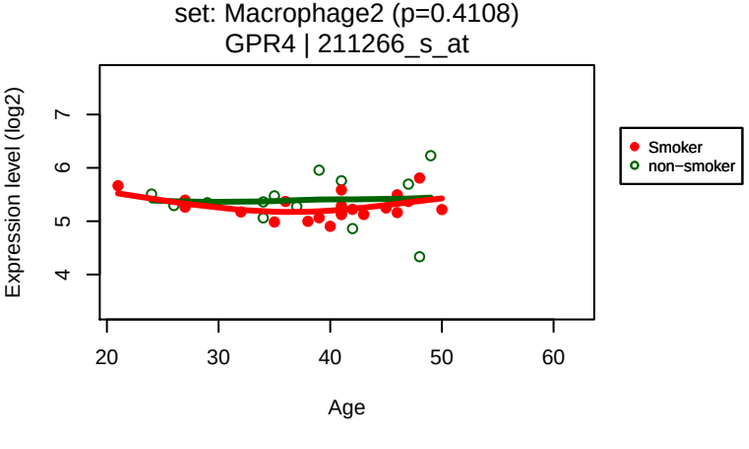

set: SmallAirway2 (p=0.1205)
GPR4 | 211266_s_at
Expression level (log2)
Age
Smoker
non−smoker

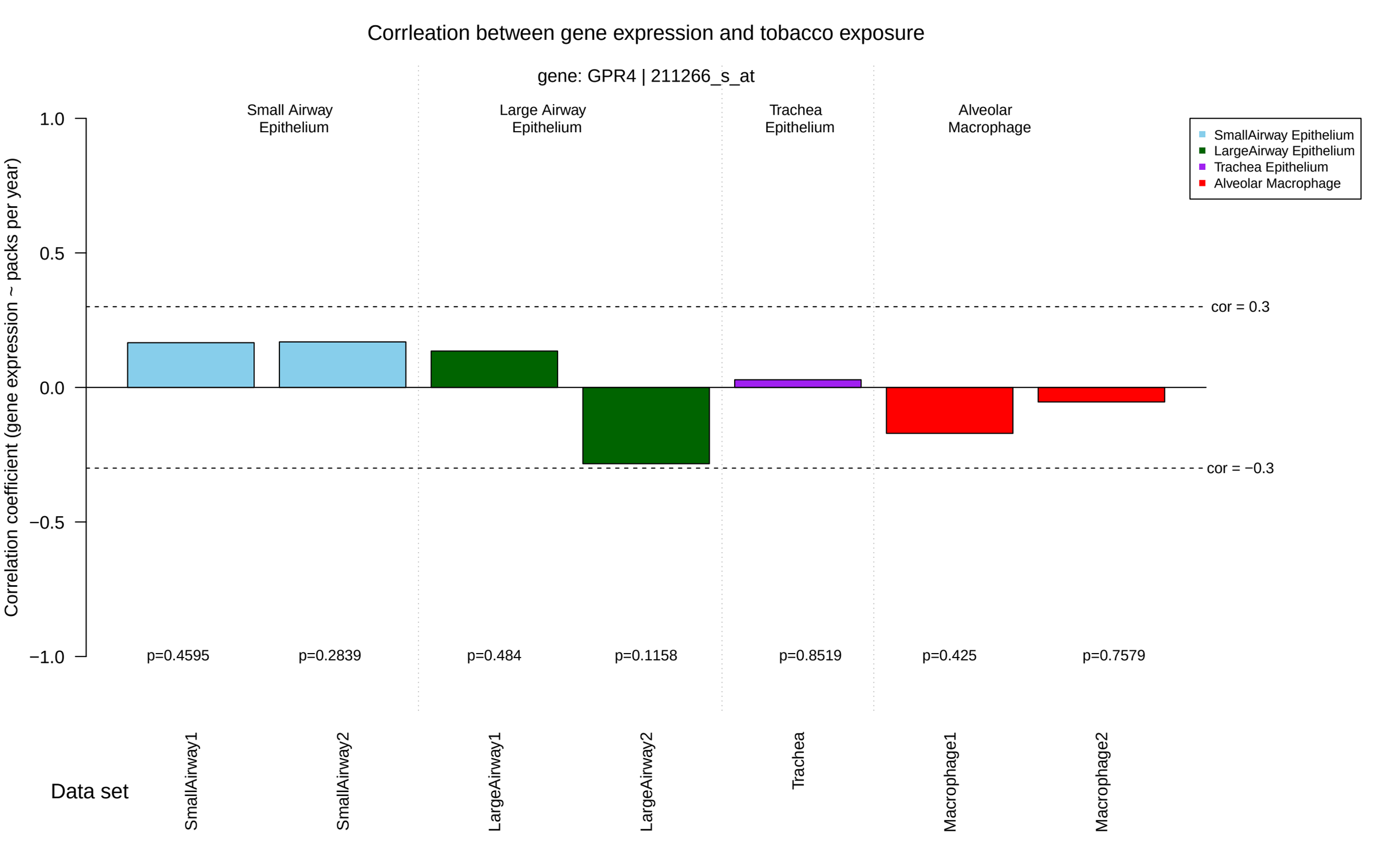

set: LargeAirway1 (p=0.0069)
GPR4 | 211266_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7209)
GPR4 | 211266_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8391)
GPR4 | 211266_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9727)
GPR4 | 211266_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4108)
GPR4 | 211266_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GPR4 | 211266_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4595
p=0.2839
p=0.484
p=0.1158
p=0.8519
p=0.425
p=0.7579
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2